PROCESS REPORT · CHEMICAL VAPOUR DEPOSITION OF 2D MATERIALS

WS_2 Synthesis by Precursor-Conversion CVD

Precursor $(NH_4)_2WS_4$, dip-coated **Furnace** single-zone tube, Ar/H_2 **Substrate** Si / SiO_2

Site Shahid Beheshti University

SUMMARY

WS_2 thin films were synthesized by a two-step thermal conversion of a dip-coated ammonium tetrathiotungstate, $(NH_4)_2WS_4$, precursor in a single-zone tube furnace under Ar/H_2. The work was carried out as a side project at Shahid Beheshti University, alongside M.Sc. research on VO_2 neuristors.

The process covered substrate preparation, precursor-solution handling, dip coating, vacuum evacuation and Ar purging, programmed two-stage annealing, and rapid cooling. No structural or optical characterization (Raman, PL, AFM, XRD) was performed, so the phase and layer number of the product are not confirmed. This report is reconstructed from a full video record of the experiment; parameters that were not recorded are given as approximate ranges and flagged as such.

Keywords: WS_2 · transition-metal dichalcogenide · ammonium tetrathiotungstate · dip coating · precursor conversion · tube furnace · Ar/H_2

30.2 mg	~400–500 °C	~800–1000 °C	~30 + 30 min
$(NH_4)_2WS_4$ weighed in the recorded run	stage 1: decomposition (approximate)	stage 2: conversion and crystallization (approximate)	hold times (recalled)

Status. The identity of the product as WS_2 is **unconfirmed**: no Raman, PL, AFM or XRD was performed.

1 Substrate preparation → 2 Precursor solution → 3 Dip coating → 4 Load and evacuate → 5 Ar purge and gas flow → 6 Two-stage anneal → 7 Rapid cooling

1 Materials and Equipment

A single-source precursor, $(NH_4)_2WS_4$, supplied both tungsten and sulfur; Ar and H_2 served as carrier and reducing gases.

Table 1. Materials and equipment.

Item	Specification	Role
Ammonium tetrathiotungstate, $(NH_4)_2WS_4$	Yellow-orange crystalline powder	Single-source W and S precursor
Solvent	Not recorded (DMF or ethylene glycol are standard for this precursor)	Dissolving the precursor for dip coating
Substrates	Si wafer pieces, likely Si/SiO_2 (thermal oxide), cut to ~1 cm scale	Growth substrate
Deionized water	M-UV-3 water purification system	Substrate rinsing
Ethanol	Analytical grade	Substrate cleaning

Analytical balance	Kern ALJ, readability 0.1 mg	Weighing precursor (30.2 mg in the recorded run)
Tube furnace	ATRA single-zone split-hinge furnace, quartz tube, stainless-steel flanges	Thermal conversion
Temperature controller	ATRA PC-11 multi-segment programmer (program Pr.05)	Two-stage ramp-and-hold profile
Mass flow controllers	Two units, readout in % full scale	Ar and H_2 flow
Vacuum system	Rotary vane pump, analog dial gauge on the inlet flange	Evacuation before growth

2 Experimental Procedure

The process ran in seven steps: substrate preparation, precursor-solution preparation, dip coating, furnace loading and evacuation, gas purging, two-stage annealing, and rapid cooling.

2.1 Substrate preparation

Si wafers were scribed and cleaved into pieces of roughly 1 cm scale. Several pieces show a yellow-green interference colour, consistent with a thermal SiO_2 layer. Cut pieces were stored wrapped in aluminium foil until use (Fig. 1a). The substrates were then cleaned with ethanol and rinsed in deionized water from the laboratory purification system (Fig. 1b).

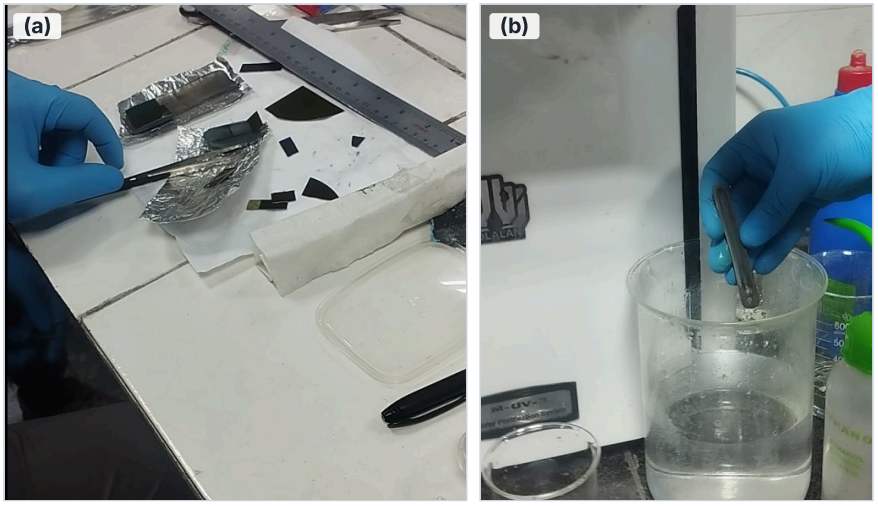


Figure 1. Substrate preparation. (a) Scribing and cleaving Si wafer pieces; cut substrates are kept wrapped in aluminium foil. (b) Rinsing a substrate in deionized water.

2.2 Precursor solution

$(NH_4)_2WS_4$ was weighed on an analytical balance; 30.2 mg was used in the recorded run (Fig. 2a). The powder was dissolved in a solvent inside a fume hood (Fig. 2b), giving the characteristic orange solution of the tetrathiotungstate anion. The solvent identity and volume were not recorded, so the solution concentration is unknown.

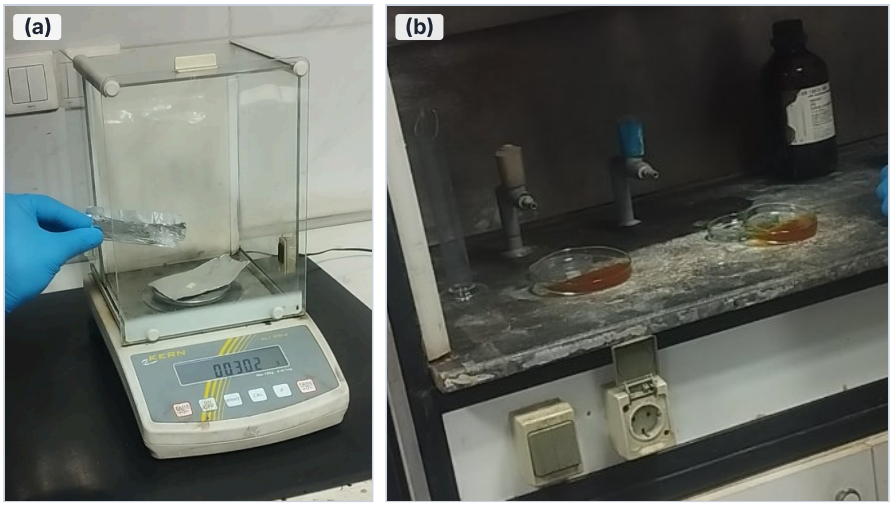


Figure 2. Precursor solution. (a) Weighing 30.2 mg of precursor on a Kern ALJ analytical balance. (b) $(NH_4)_2WS_4$ solution in Petri dishes inside the fume hood.

2.3 Dip coating

Cleaned substrates were immersed in the precursor solution and withdrawn, leaving a thin precursor film on the surface. Undissolved yellow crystallites visible in the beaker indicate the solution was close to saturation (Fig. 3a). Dip duration and withdrawal speed were not recorded.

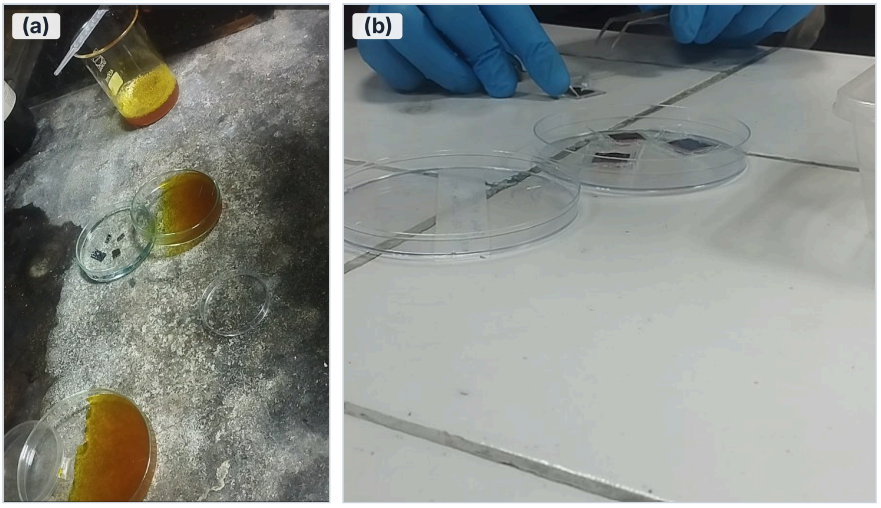


Figure 3. Dip coating. (a) Precursor solution with undissolved crystallites (top) and coated substrates in a Petri dish (centre-left). (b) Precursor-coated substrates transferred to a clean Petri dish before loading.

2.4 Furnace loading and evacuation

Coated substrates were loaded into the quartz tube of a single-zone split-hinge tube furnace (Fig. 4a). The tube was evacuated with a rotary vane pump; pressure was monitored on an analog dial gauge mounted on the gas-inlet flange (Fig. 4b). The section of tube between the inlet flange and the furnace body was wrapped in aluminium foil.



Figure 4. Furnace and vacuum line. (a) Quartz tube seated in the opened split-hinge furnace, showing the ceramic-fibre insulation and heating elements. (b) Gas-inlet flange with analog pressure gauge; the upstream tube section is wrapped in aluminium foil.

2.5 Gas purging and flow

After evacuation, the tube was purged with Ar. Two mass flow controllers delivered Ar and, most likely, H_2 during annealing; H_2 acts as the reducing agent that strips excess sulfur from the intermediate WS_3. The controllers read in percent of full scale, and the setpoints and full-scale ranges were not recorded, so absolute flow rates in sccm are unknown.

2.6 Two-stage thermal program

The annealing ran as program Pr.05 on the ATRA PC-11 controller (Fig. 5b). The exact setpoints were not recorded; the temperatures in Table 2 are the standard ranges for this precursor route, and the hold times are as recalled.

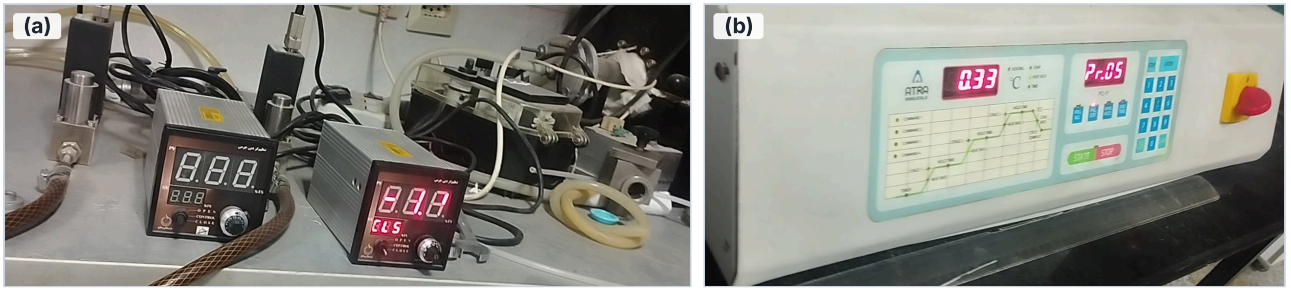


Figure 5. Gas and temperature control. (a) The two mass flow controllers (readout in % full scale) used for Ar and H_2. (b) ATRA PC-11 multi-segment temperature controller with program Pr.05 selected.

Table 2. Two-stage thermal program (program Pr.05).

Stage	Temperature (°C)	Hold (min)	Purpose
1 — decomposition	~400–500 **(approximate, not recorded)**	~30 (recalled)	Decompose $(NH_4)_2WS_4$ to WS_3, release NH_3 and H_2S
2 — conversion and crystallization	~800–1000 **(approximate, not recorded)**	~30 (recalled)	Reduce WS_3 to WS_2 and crystallize the film

2.7 Rapid cooling

At the end of the second hold, the split-hinge furnace was opened to cool the tube rapidly under gas flow. Fast cooling limits sulfur loss and further grain coarsening at high temperature.

3 Conversion Chemistry

The precursor converts to WS_2 in two thermally separated reactions, which is why the program uses two holds.

Stage 1. $(NH_4)_2WS_4$ decomposes to amorphous tungsten trisulfide, releasing ammonia and hydrogen sulfide:

$$(NH_4)_2WS_4 \xrightarrow{\sim 300\text{–}500\ ^\circ C} WS_3 + 2\ NH_3 + H_2S$$

Stage 2. WS_3 loses its excess sulfur. H_2 in the gas stream drives this reduction and removes sulfur as H_2S, while the high temperature crystallizes the 2H-WS_2 lattice:

$$WS_3 + H_2 \xrightarrow{\sim 800\text{–}1000\ ^\circ C} WS_2 + H_2S$$

Because the precursor already contains sulfur in a 1:4 W:S ratio, no separate sulfur source is strictly required on this route; whether additional sulfur powder was placed upstream was not recorded.

4 Outcome and Limitations

The run produced uniformly dark films on the Si substrates, but their identity as WS_2 is unconfirmed because no characterization was carried out.



Figure 6. Samples after annealing, stored in a Petri dish and a sealed sample bag.

Parameters not recorded during the experiment are listed in Table 3.

Table 3. Parameters not recorded.

Parameter	Status
Solvent and solution concentration	Not recorded
Dip time and withdrawal speed	Not recorded
Stage 1 and stage 2 setpoints	Approximate ranges only
Heating ramp rate	Not recorded
Ar and H_2 flow rates (sccm)	Not recorded
Working pressure during annealing	Not recorded
Additional sulfur source	Not recorded

Recommended characterization. The standard checks that would confirm the result are Raman spectroscopy (E^1_{2g}/2LA mode near 350–356 cm^{-1} and A_{1g} near 418 cm^{-1}, with their separation indicating layer number), photoluminescence (direct-gap emission near 620–630 nm for a monolayer), AFM step height for thickness, and XPS for the W:S ratio and residual oxide.

5 Skills Demonstrated

- **Substrate preparation:** wafer scribing and cleaving, solvent and DI-water cleaning
- **Precursor chemistry:** handling and weighing a sulfide precursor at 0.1 mg precision, solution preparation in a fume hood
- **Thin-film deposition** from solution by dip coating
- **Vacuum practice:** rotary-pump evacuation, inert-gas purging, pressure monitoring
- **Gas handling** with mass flow controllers, including H_2 as a reducing gas
- **Thermal processing:** programming multi-segment ramp-and-hold profiles on a tube-furnace controller
- **Process design** for a two-step decomposition and reduction route to a 2D transition-metal dichalcogenide